

1. Flowchart & Pseudocode + Installation

Time Stamps :

1. How to study this series?
2. How to solve problems?
3. Flowchart
4. Pseudocode
5. Practice Qs1
6. Practice Qs2
7. Practice Qs3
8. Practice Qs4
9. Practice Qs5
10. Homework Problems
11. How code runs?
12. What to Install?

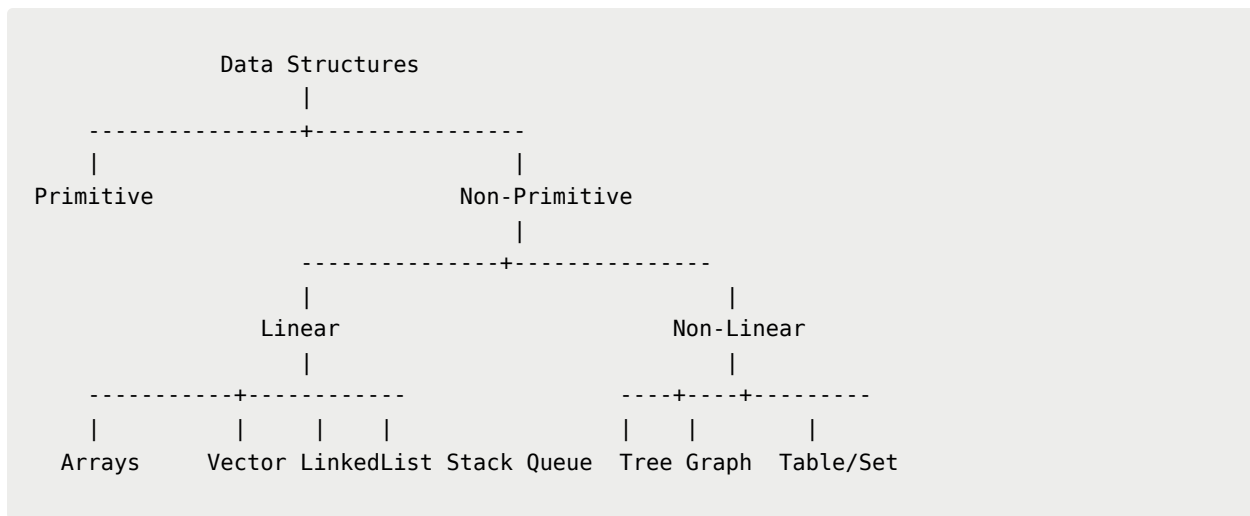
1. How to study this series

- i. About series
 - Indepth & Simple
 - Good DSA concept, helps us good problem solve as a SE.
 - Resolve 3 problems: 0 → hero, Objective quality DSA series, Practice questions

ii. Why to learn DSA

- For SDE & Tech Roles
- Companies ask coding DSA questions in interviews & coding tests and to solve that questions unke liye we must know DSA.

iii. Data Structures



Also Cover: Algorithms & Approches (DP, Greedy Algo's, 2 Pointer Approach)

iv. C++ & Topics

- C++: 1st start with a programming lang. → C++
- Topics: Flow & Psue, Var DT Ope, Conditional Statements, loops, Functions, Arrays, Pointers, BNS

v. Notes:

- Don't need to make st lec notes.
- Arrays, pointers, & Binary no. me itne bar pdh lenge, ki ABCD ki treh yad hone wale hai.

vi. How to Practice

- class notes, hw, and aage lectures dekhte dekhte to practice ho hi jayenge.
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2. How to solve a problem?

- Problem: Find Sum of 2 no.s a & b:
 - Understand → Input → Solution → Code.
-

3. Flowchart

- Flowchart: Diagram of a solution
 - Components: start, exit (circle) → in, op (parallelogram) → Process (rectangle) → decision block (diamond)
 - Note: FC just for logic building, Not imp in interview, pseudo is good to know. But in interview we will write direct code.
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4. Pseudo-code

- Defination: General logic of solution.
 - Eng. like lang, Anyone can understand, Irrespective of code, Ignore St, exit is optional
 - //jaise hmne FC and sudo code dekha bs isi logic ko hm code me convert krte hai.
-

5. Q1. Area of Sq.

FC

Start (in circle)

Input a (parallelogram)

area = a * a

print area

exit

Pseudo code (pseudo means nkli)

1. Input a

2. area = a * a

3. print area

4. exit

6. Q2. Min of 2 no.s (In notebook)

7. Q3. Even/ Odd

```
int n
cin >> n;
if(n%2==0) {
    cout << "even" << endl;;
} else {
    cout << "odd" << endl;
}

return 0;
```

8. Q4. SN natural no.s 1 to N

- Direct Formula: $sn = n*(n+1)/2$

```

#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> nums = {1, 2, 3, 4, 5};
    int count = 0;
    for(int i = 0; i < nums.size(); i++) {
        count += nums[i];
    }

    cout << count << "\n"; //backslash n (newline character)

    return 0;
}

//Note: main me return count ni krna. cout hi krna hai. bcoz
return will compile, run, but will not
//shown on screen. Screen pe print show hota hai. More unders
tanding in Lec 5 Functions.

//Note: cpp code is more easier than pseudocode.

```

9. Q5. Prime or not (see in Notebook)

- 2 factor → (1, n), n means khud wo no.
- 2 is only a even prime no. So it is a special prime no.

10. HW

Write FC & Pseudocode for:

- a. SI
- b. Max of 2 no.s
- c. Factorial/ product of no. 1 to N
- d. given age, find should get driving license or not. ← similar question to can take vote or not.

```
// c. Factorial
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> nums;
    int prod = 1;

    for(int i=0; i<nums.size(); i++) {
        prod *= nums[i];
    }

    cout << prod << "\n";

    return 0;
}
```

11. How does code runs?

- c++ code → Translator (c++ compiler, needs to install) → 0/1 (executable file)
→ output

//executable file means: A file that can run.

12. What to install?

- What to download? a. VS Code b. Set up C++
- VS Code: is an editor. very good software we can use for DSA, WEB dev and more.

Note 1:

- Error can come while install vs code. which is completely normal things. But we don't need to hesitate that error aa gya, mene chor diya. Bcoz we are learning DSA and DSA se bdkr we are learning to solve the problem.
- If any error comes, then google it. Bcoz internet has almost everything solution.
- And jo software we are going to install. That are install by lakhs of people before us. So, unke solution hme internet pr mil jate hai.

Note 2:

- Browser Based Website [Not recommended to use for dsa]
 - Bcoz we are learning DSA for long term not only for 1-2 months.
 - VS code will help us in long term.
 - To be a programmer, software engineer. We must know to install a software on our system.
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At last

Option	What to do	Why
1	Comment: Lecture, Date, Completed	Sense of achievement as a student
2	Share didi's Twitter link in the description (Shradha Khapra). You can tag mam on Twitter.	Marking small small ticks gives a feeling of happiness in the long term. Learning in public. → twitter a/c to track students performance.

Complete :)